Greecon
SH.P.K

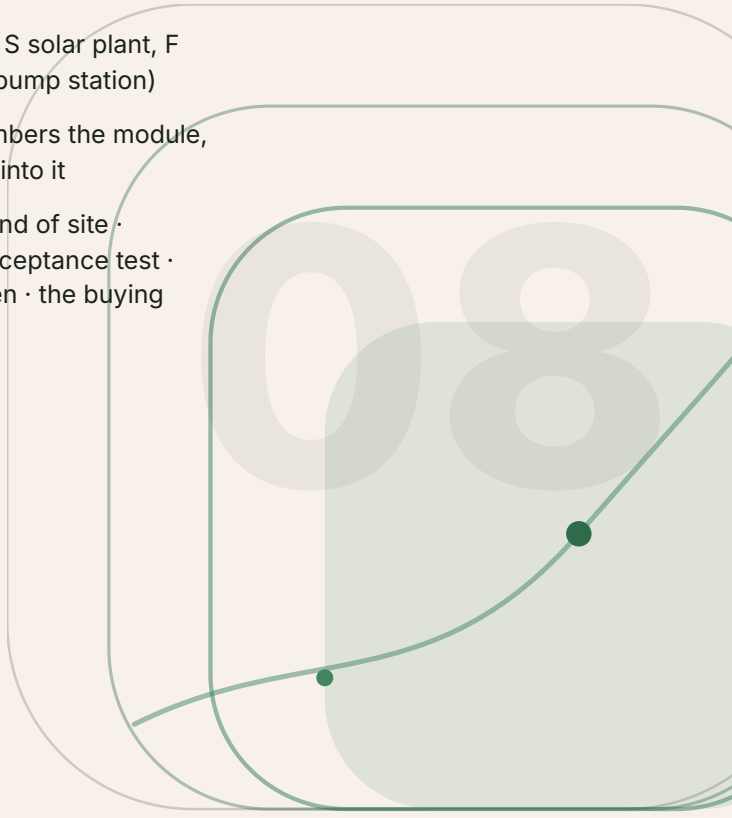
08

HARDWARE SHEET HW08 · APPROVED MODEL

4 to 20 mA module: Waveshare Analog Input 8CH

The module that reads the level, pressure and flow transmitters and puts their signals on the data cable. The platform calls it the Water I/O module.

QUALIFIES	Approved model. The Waveshare Modbus RTU Analog Input 8CH, in the version that reads current. The version that reads voltage is a different part.
USED IN	Kits H, F and W (H hotel or guesthouse, S solar plant, F farm or greenhouse, W water tank and pump station)
FOR	The project manager who buys and numbers the module, and the electrician who wires the loops into it
COVERS	How it qualifies · what it does at each kind of site · settings · fitting · in the platform · the acceptance test · what goes wrong · what is not yet proven · the buying record



Version 1.0 · 10 November 2026

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HW08

4 to 20 mA module: Waveshare Analog Input 8CH

The level, pressure and flow transmitters each send a current between 4 and 20 mA down two wires. This module reads those currents and answers the box with numbers, on the same data cable as the meters. The platform calls it the **Water I/O module**. It is an approved model, so the box finds it, and the only thing a person is asked for is the range printed on each transmitter's label.

At a glance

WHAT IT DOES	Reads the 4 to 20 mA signals of the level, pressure and flow transmitters (inputs 1, 2 and 3) and answers the box on the same RS-485 cable as the meters, as unit 3. It is left in its default 0 to 20 mA mode, so that a wire that has come off reads zero and Greecon can say so.
HOW IT QUALIFIES	**Approved model.** The Waveshare Modbus RTU Analog Input 8CH, in the version that reads current. The version that reads voltage is a different part.
MAKE AND MODEL	Waveshare Modbus RTU Analog Input 8CH (current version, the one that reads current), 8 inputs, 4 to 20 mA, 7 to 36 V DC
NUMBER ON THE CABLE	3. Set with the maker's setting tool, at the office (it is delivered as 1).
FITTED BY	The project manager mounts it in the enclosure and joins it to the data cable. The electrician wires the 4 to 20 mA loops into its inputs.
PLANNING ESTIMATE (EUR)	30 to 60. Not a quote.
USED IN	Kits H, F and W. The table under *At each kind of site* says what it does at each.

How it qualifies

It is an approved model: the make and the model are named.

It qualifies if:

- ☐ It is the Waveshare **Modbus RTU Analog Input 8CH**, in its current version: eight inputs, on RS-485, powered from 7 to 36 V DC.
- ☐ It is the version that **reads current**. The version that reads voltage is a different part, and does not qualify.
- ☐ The maker's setting tool for it is on the office laptop, with a USB-to-RS-485 adapter: its number is set by a command, not by a switch.

STOP

Do not buy another make of analog input module, or the voltage version of this one. The box recognises this module by a register that holds its own number, and reads its inputs as microamps: another module answers differently, or not at all.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	IN THE KIT	1	One. Input 1 takes the tank's level transmitter, input 2 the pipe's pressure transmitter, input 3 the flow meter if there is one. It gives Tank level, Line pressure and Pump flow.
S Solar plant with grid connection	NOT USED	None	A solar plant has no water side.
F Farm or greenhouse with irrigation	IN THE KIT	1	One, for the tank and the pipe behind the irrigation. The flow meter (input 3) only if there is a pump to watch.
W Water tank and pump station	IN THE KIT	1	One, the centre of the kit: level on input 1, pressure on input 2, the flow meter on input 3 if it is fitted.

What Greecon reads from it

Each input is one input register that holds the current in microamps: input 1 is register 0, input 2 is register 1, input 3 is register 2. A 4 to 20 mA transmitter reads 4000 to 20000. Greecon turns that into the reading from the range on the transmitter's label.

Input	Transmitter	Reading in Greecon	The label gives	Standard alert
1	Level (sheet HW10)	**Tank level**, %	The range, in metres or bar, and how tall the tank is	Critical below 15 % (Act now); Low below 35 % (Check today)
2	Pressure (sheet HW11)	**Line pressure**, bar	The top of the range, in bar	None
3	Flow (sheet HW12), if fitted	**Pump flow**, litres a minute	The top of the range, in litres a minute	No flow at or below 0.2 litres a minute (Act now), only when the pump's running state is read too

Its inputs are left in their default 0 to 20 mA mode, on purpose. A wire that has come off then reads 0, which is below the 4 mA of an empty tank, so Greecon can see it and say so (*cannot be a tank level*). In the module's own 4 to 20 mA mode a broken wire might read like an empty tank.

Settings, and where they are made

Setting	Value	Where
Its number	**3.** It is delivered as 1, and its number is kept in a register: there is no switch. Set it with the maker's setting tool and a USB-to-RS-485 adapter, with no other device on the cable	At the office
Speed	9600 baud, 8 data bits, no parity, 1 stop bit, which is how it is delivered. Check it	At the office
Input mode	Left at the default, 0 to 20 mA. Do not change it	Nowhere

Label the module with its number, and photograph the label.

Fitting

PROJECT MANAGER mounts the module on the DIN rail in the enclosure, powers it from the 24 V supply and joins it to the data cable. ELECTRICIAN wires the transmitters' loops into its inputs.

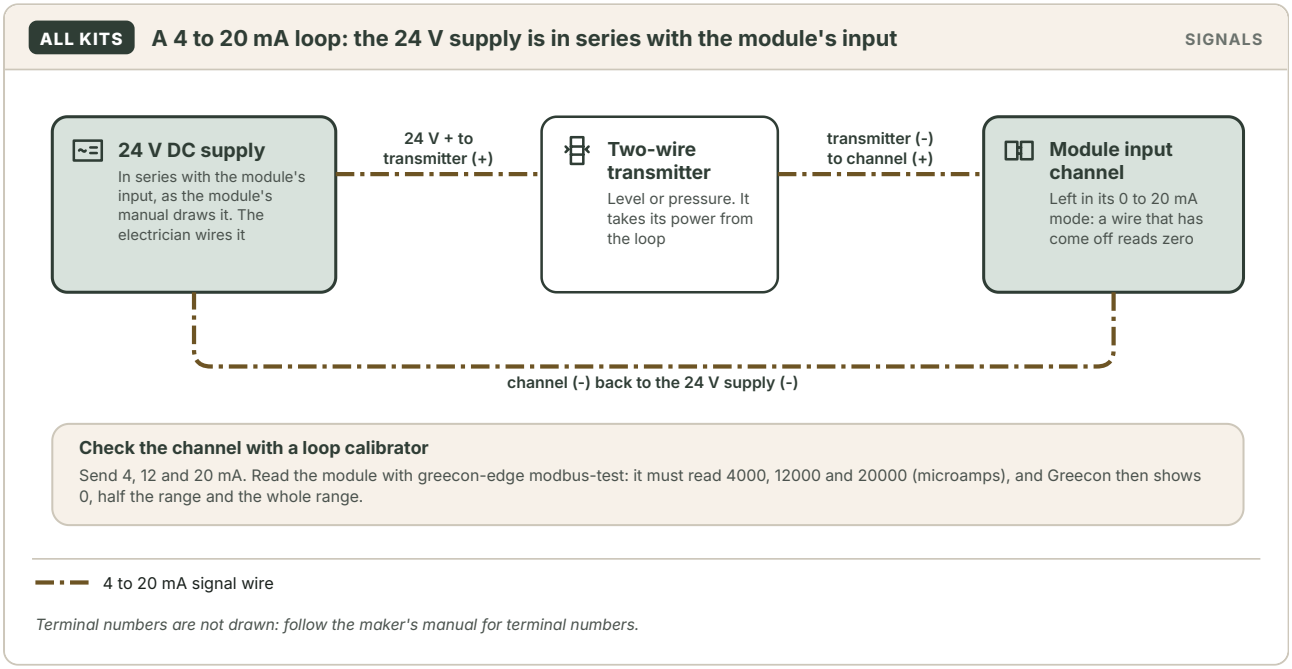


Figure 4-3. A 4 to 20 mA loop: the 24 V supply in series with the module's input.

- ☐ **Each transmitter has its own input:** level on input 1, pressure on input 2, the flow meter on input 3.
- ☐ **A two-wire transmitter takes its power from the loop,** so each loop has the 24 V supply in series with the module's input, wired as the module's manual draws it. The maker does not say that the module powers the loop, so the loop has a supply of its own.
- ☐ Number every wire at both ends, and photograph the terminals before the enclosure is closed.

In the platform

- **Finding it.** In step 4 of the setup list, **Where each device is on the network**, press **Find equipment**. The box recognises the module by the register that holds its own number, and by its inputs. Found at number 3, its address is filled in: *unit 3: a 4 to 20 mA input module, reading 12.0 mA, 9.0 mA, 5.1 mA*.
- **The ranges.** In step 5, **Which register holds which reading**, the page asks one thing for each transmitter, in the words of its label: the **Range on the label** and the **Unit on the label** (metres or bar) of the level transmitter and **How tall the tank is inside**; the **Top of the range** of the pressure transmitter, and of the flow meter. Press **Save the ranges**. Greecon works out the register and the scale: nobody types either.
- **A transmitter the site does not have** (no flow meter): tick **Not fitted at this site**. The page asks first, because the reading and its history are removed.

Acceptance test

At the office, on the desk:

- ☐ The module is numbered 3, alone on the cable, and labelled.
- ☐ Behind the converter, the box finds it at number 3 (`greecon-edge find --host 192.168.1.50`).
- ☐ With a loop calibrator in place of a transmitter (a hand-held that sends 4, 12 and 20 mA), the module reads 4000, 12000 and 20000 on that input. Address 0 is input 1, 1 is input 2, 2 is input 3:

```
greecon-edge modbus-test --host 192.168.1.50 --unit 3 --type input --address 0
```

- ☐ With nothing connected to an input, it reads 0.

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
4 to 20 mA channels. With a loop calibrator, send 4, 12 and 20 mA into each input that is used.	The reading at the bottom, the middle and the top of its range			☐ Pass ☐ Fail

On the tank's input, the top of the range reads more than 100 when the transmitter is longer than the tank (125 for a 0 to 5 m transmitter in a tank 4 m tall), and Greecon may call that impossible while you test. That is the test, not a fault.

What goes wrong

What you see	Likely cause	What to do
A reading below zero (*-31.2 cannot be a tank level*)	No current in the loop: a wire has come off, or the loop has lost its 24 V supply. The module reads 0	Read the input from the box. 0 means no current. The electrician checks the loop and its supply

What you see	Likely cause	What to do
A reading far too high or too low	The range, or the tank's height, was given wrongly; or a label in bar was entered as metres (it then reads about ten times too low)	Read the label again and give the ranges again
The right number on the wrong reading	Two transmitters on each other's inputs	Level on input 1, pressure on input 2, flow on input 3
Nothing that looks like a 4 to 20 mA input module answers at unit 3	The module has no power, was left at number 1, or its A and B wires	Check its power and its two data wires. Read its label. A module that was not numbered at the office goes back there, or the spare that was numbered there is fitted

More is in the runbooks (Project Manager Field Manual, PDF 02): R15 (a device is not found) and R18 (a reading is impossible after the ranges were given).

Not yet proven

- **This software has not yet read a real one of these modules.** Its registers (the inputs at input registers 0 to 7, in microamps; its number at holding register 0x4000) are from the maker's published pages, read again on 31 October 2026, and everything is tested against a pretend module that answers as those pages say. The first real kit is rehearsed on a desk, and anything that differs from this sheet is written down and reported to Greecon.
- The prices are planning figures from shop listings seen in October 2026, not quotes.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	